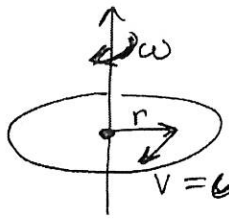
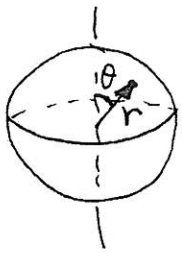
P5.6

a)



$v = \omega r \Rightarrow K = \sigma v = \sigma \omega r$

b)



$\vec{J} = \rho \vec{v}$

$\vec{\omega} = \omega \hat{z} = \omega(\cos\theta\, \hat{r} - \sin\theta\, \hat{\theta})$

$\vec{v} = \vec{\omega} \times \vec{r}$

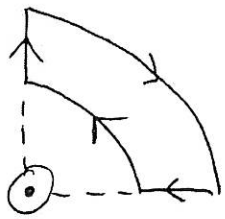
$\Rightarrow \vec{v} = -\omega \sin\theta\, r\, \underbrace{\hat{\theta} \times \hat{r}}_{-\hat{\varphi}} = \omega r \sin\theta\, \hat{\varphi}$

$\rho = \dfrac{Q}{\frac{4\pi}{3} R^3}$

$$\vec{J} = \frac{Q}{\frac{4\pi}{3} R^3}\, \omega r \sin\theta\, \hat{\varphi}$$

P5.9

a)



$$\vec{B}(\vec{r}) = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l} \times (\hat{r} - \hat{r}')}{|\vec{r} - \vec{r}'|^2}$$

$d\vec{l} \times \hat{r} \Rightarrow \hat{z}$ = normal vector out of page

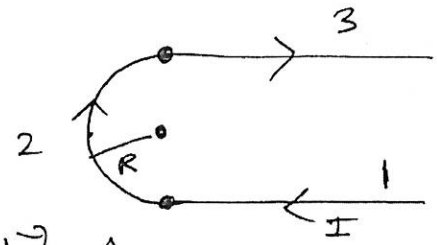
$$\vec{B}_a = \frac{\mu_0 I}{4\pi}\, \hat{z} \int_0^{\pi/2} \frac{a\, d\varphi}{a^2} = \frac{\mu_0 I}{8a}\, \hat{z}$$

$$\vec{B}_b = \frac{\mu_0 I}{4\pi}\, \hat{z} \int_{\pi/2}^{0} \frac{b\, d\varphi}{b^2} = -\frac{\mu_0 I}{8b}\, \hat{z}$$

Straight segments do not contribute.

$$\vec{B} = \frac{\mu_0 I}{8} \left(\frac{1}{a} - \frac{1}{b} \right) \hat{z}$$

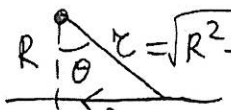
f)



$\vec{B}$ is into the page.

$d\vec{l} \times \hat{r}$ is into the page $= (-\hat{z}) \cdot \sin\left(\frac{\pi}{2} + \theta\right) = (-\hat{z}) \cos\theta$

Segment 1: $\vec{B}_1 = \frac{\mu_0 I}{4\pi}(-\hat{z}) \int_{-\pi/2}^{0} \cos\theta \frac{R\,d\theta}{\cos^2\theta} \frac{1}{(R/\cos\theta)^2} =$



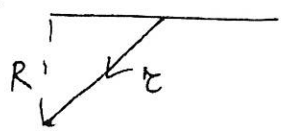
$r = \sqrt{R^2 + l^2} = \frac{R}{\cos\theta}$

$l = R\tan\theta \Rightarrow dl = \frac{R\,d\theta}{\cos^2\theta}$

$= \frac{\mu_0 I}{4\pi R} \sin\theta \Big|_{-\pi/2}^{0} (-\hat{z})$

$= \frac{\mu_0 I}{4\pi R}(-\hat{z}).$

Segment 3:



$\vec{B}_3 = \vec{B}_1$

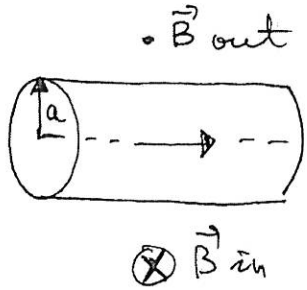
segment 2: $\vec{B}_2 = \frac{\mu_0 I}{4\pi}(-\hat{z}) \int_0^{\pi} \frac{R\,d\varphi}{R^2} = \frac{\mu_0 I}{4R}(-\hat{z})$

$\Rightarrow \vec{B} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3 = (-\hat{z})\left(\frac{2\mu_0 I}{4\pi R} + \frac{\mu_0 I}{4R}\right)$

$= (-\hat{z}) \frac{\mu_0 I}{4R}\left(1 + \frac{2}{\pi}\right).$

P5.13

a)



$\vec{J}$ = const on the outside surface of the wire

$\Rightarrow$ we have surface current $K = \frac{I}{2\pi a}$

By symmetry $\vec{B}$ circles around the wire.

$$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc} = \mu_0 \begin{cases} 0 & \text{for } r<a \\ I & \text{for } r>a \end{cases}$$

$$\| $$

$$B\cdot 2\pi r$$

$$\Rightarrow B = \begin{cases} 0 & \text{for } r<a \\ \frac{\mu_0 I}{2\pi r}\hat{\varphi} & \text{for } r>a \end{cases}$$

b) $J = ks$ (s = distance from the wire axis)

total $I = \int J\,da = \int_0^a ks\cdot 2\pi s\,ds = 2\pi k\int_0^a s^2 ds = \frac{2\pi k}{3}a^3$

$$\Rightarrow k = \frac{3I}{2\pi a^3}.$$

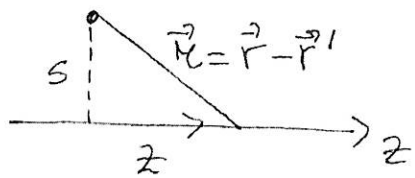
$$\Rightarrow B\cdot 2\pi s = \cancel{\mu_0 \int_0^s dr\, kr = \mu_0 k \frac{s^2}{2}}\ \mu_0 I_{enc}$$

$$I_{enclosed} = \int_0^s kr\cdot 2\pi r\,dr = k\,2\pi\frac{s^3}{3}.$$

$$\Rightarrow B(s) = \mu_0 k\frac{s^2}{3} = \frac{\mu_0 I}{2\pi a^3}s^2 \text{ for } s<a.$$

$$B(s) = \frac{\mu_0 I}{2\pi s} \text{ for } s>a \text{ of course.}$$

P5.22 $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\int \frac{\vec{J}(\vec{r}\,')\,dV'}{|\vec{r}-\vec{r}\,'|} = \frac{\mu_0 I\hat{z}}{4\pi}\int \frac{dz}{|\vec{r}-\vec{r}\,'|}$



$$\vec{A} = \frac{\mu_0 I\hat{z}}{4\pi}\int_{z_1}^{z_2}\frac{dz}{\sqrt{s^2+z^2}} = \frac{\mu_0 I\hat{z}}{4\pi}\ln\left(z+\sqrt{s^2+z^2}\right)\Big|_{z_1}^{z_2} =$$

$$= \frac{\mu_0 I\hat{z}}{4\pi}\ln\left(\frac{z_2+\sqrt{z_2^2+s^2}}{z_1+\sqrt{z_1^2+s^2}}\right); \quad \vec{A} = A_z\hat{z}.$$

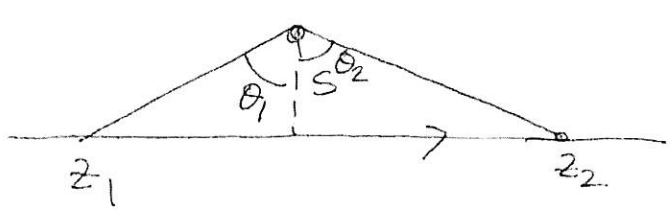
Cylindrical coordinates $\vec{B} = \vec{\nabla}\times\vec{A} = \frac{1}{s}\underbrace{\left(\frac{\partial A_z}{\partial\varphi}\right)}_{=0}\hat{s} - \frac{\partial A_z}{\partial s}\hat{\varphi}$

$$\frac{\partial A_z}{\partial s} = \frac{\mu_0 I}{4\pi}\left(\frac{1}{z_2+\sqrt{z_2^2+s^2}}\,\frac{2s}{2\sqrt{z_2^2+s^2}} - \frac{1}{z_1+\sqrt{z_1^2+s^2}}\,\frac{s}{\sqrt{z_1^2+s^2}}\right)$$

$$= \frac{\mu_0 I s}{4\pi}\left(\frac{z_2-\sqrt{z_2^2+s^2}}{z_2^2-(z_2^2+s^2)}\,\frac{1}{\sqrt{z_2^2+s^2}} - \frac{z_1-\sqrt{z_1^2+s^2}}{z_1^2-(z_1^2+s^2)}\,\frac{1}{\sqrt{z_1^2+s^2}}\right)$$

$$= -\frac{\mu_0 I s}{4\pi s^2}\left(\frac{z_2}{\sqrt{z_2^2+s^2}} - \frac{z_1}{\sqrt{z_1^2+s^2}}\right)$$

$$\Rightarrow \vec{B} = \frac{\mu_0 I}{4\pi s}\left(\sin\theta_2 - \sin\theta_1\right)$$



P5.24 $\vec{B} = \text{const}$. Check $\vec{A} = -\frac{1}{2}(\vec{r} \times \vec{B})$

$$\vec{\nabla} \cdot \vec{A} = \left(-\frac{1}{2}\right) \vec{\nabla} \cdot (\vec{r} \times \vec{B}) = -\frac{1}{2}\left(\vec{B} \cdot \vec{\nabla} \times \vec{r} - \vec{r} \cdot \vec{\nabla} \times \vec{B}\right)$$

(product rule 6)

$$\vec{\nabla} \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = 0, \text{ and } \vec{\nabla} \times \vec{B} = 0 \text{ as } \vec{B} = \text{const}$$

$$\Rightarrow \vec{\nabla} \cdot \vec{A} = 0 \quad \underline{\text{or}}$$

$$\vec{B} = \vec{\nabla} \times \vec{A} = \left(-\frac{1}{2}\right) \vec{\nabla} \times (\vec{r} \times \vec{B}) = -\frac{1}{2}\Big((\vec{B} \cdot \vec{\nabla})\vec{r} - (\vec{r} \cdot \vec{\nabla})\vec{B}$$

(rule 8) $\quad + \vec{r}(\vec{\nabla} \cdot \vec{B}) - \vec{B}(\vec{\nabla} \cdot \vec{r})\Big)$

$$= \frac{1}{2}\left\{\left(B_x \frac{\partial}{\partial x} + B_y \frac{\partial}{\partial y} + B_z \frac{\partial}{\partial z}\right)(x\hat{x} + y\hat{y} + z\hat{z})\right.$$

$$\left. - \vec{B}\left(\frac{\partial}{\partial x} x + \frac{\partial}{\partial y} y + \frac{\partial}{\partial z} z\right)\right\} =$$

$$= -\frac{1}{2}\left\{\underbrace{B_x \hat{x} + B_y \hat{y} + B_z \hat{z}}_{\vec{B}} - 3\vec{B}\right\} = -\frac{1}{2}(-2\vec{B}) = \vec{B}.$$

It works but is not unique.